MAINS MASTER PROGRAM (MMP) 2024
SCIENCE AND TECHNOLOGY-4

1. TABLE OF CONTENTS

1. Table of Contents 0

2. Nuclear Science and Technology 1

1) Nuclear Fusion 1

2) Recent Progress towards harnessing Nuclear Fusion Energy 2

3) India's Three-Phase Nuclear Power Program... 3

4) Civil Liability for Nuclear Damages (CLND) Act, 2010.. 4

5) Nuclear Energy and Energy Security 5

6) Use of Nuclear Radiation tECHNOLOGY FOR PROVIDING BETTER QUALITY OF LIFE.... 6

7) Important Scientists and their Contributions: Homi Jehangir Bhabha.........10

8) Radiocarbon Dating.............................10

3. Computer, IT and ELectronics 12

1) 5G 12

2) 6G...14

A) Bharat 6G Alliance (B6GA)..............................14

B) Bharat 6G MisSion..15

3) Web Browsers 15

A) Request and Response 15

B) Deconstructing The Response 15

C) Rendering 16

D) Managing Data 16

E) Security 16

4) Web 3.0 17

5) Blockchain Technology.........................18

6) National Strategy on Blockchain 19

7) Supercomputers 20

8) Supercomputing in India 21

9) Quantum Computers 22

10) Internet of Things (IoT)....................24

11) Big Data 25

12) Note: 26

2. NUCLEAR SCIENCE AND TECHNOLOGY

1) NUCLEAR FUSION

- **Practice Questions**:
  » What is Nuclear Fusion Energy? What are its advantages of Nuclear Fission energy? [10 marks, 150 words]

- **Introduction**:
  » In a nuclear fusion reaction, two or more nuclei fuse together to from one heavier nucleus. In the process some mass is lost which gets converted into energy as per Einstein's Mass Energy Equivalence ($E = mc^2$). This is known as nuclear fusion energy. For e.g:

$$4\,{}^{1}_{1}H \rightarrow {}^{4}_{2}He + 2\,{}^{0}_{+1}e$$

  » It is the source of energy in all the stars. But, so far, humankind has not been able to harness this reaction for civilian energy production. It is because fusion reaction requires **very high temperature** and **pressure**.

- **Still, scientists all across the world are working to develop technologies to harness fusion energy** for electricity production. This is because it has several advantages over nuclear fission energy reaction:
  1) **Raw material** (hydrogen atom) is easily available in the universe unlike Uranium needed for fission.
  2) **Efficiency**: Hydrogen fusion reaction is one of the most efficient energy production methods in the universe. It produces four times more energy than a standard Uranium based fission reaction.
  3) **Safety**: Fusion energy reactions are inherently safer as they can't undergo uncontrolled chain reactions which may be the case with fusion.
  4) **Environment Friendly**: The fusion reaction is clean and green route to produce energy and doesn't involve any remnant waste products. Whereas Fission reaction produces a lot of radioactive waste whose disposal remains a challenge.
  5) **Long term energy security**: As raw material can be obtained easily (by electrolysis of water), this energy type can ensure energy security for every nation of the world.

- Therefore, scientists across the world are working to make controlled fusion reactions successful. International Thermonuclear Experimental Reactor (ITER) are working on the Magnetic confinement method and USA's National Ignition facility is working on the Inertial Confinement Method.

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2) RECENT PROGRESS TOWARDS HARNESSING NUCLEAR FUSION ENERGY

Practice Question: "Use of fusion process for generating electricity at a commercial scale is decades away, but the latest experiment by US scientists is still a big deal. Elaborate. How is their method different from the one being used by ITER? [15 marks, 250 words]

Ans:

So far, no country in the world has been able to develop functional thermonuclear energy reactor. It is because the following conditions must be fulfilled for reactions to take place:

1) **Very high temperature** (around 15 million degree C)
2) **Sufficient Plasma particle density** (to increase the likelihood that collision occurs)
3) **Sufficient confinement time** (to hold the plasma, which has the propensity to expand, within a defined volume)

But, in recent years the National Ignition Facility (NIF) of USA have conducted experiments which achieved **fusion ignition** and they were able to achieve a gain of **2** when compared to the energy of income laser. They used an **Inertial Confinement method**.

- In NIF's set up, high power lasers fire pulses at a 2 mm wide capsule inside a 1-cm long cylinder called **hohlraum**, in less than 10 billionth of a second. This capsule holds (Deterium-Tritium). A short window is created where nuclei can fuse.

This is the first time a gain of nearly 2 has been achieved on an fusion energy experiment on earth. It is being considered the most impressive feat of the 21st century and is an engineering marvel beyond belief.

This is despite the fact that use of fusion process for **generating electricity** at commercial scale may still be **decades away**. A gain of more than 100 will be required and NIF's experiment to powerplant transition is till not well understood.

Differences between NIF and ITER methods:

National Ignition Facility	**International Thermonuclear Experimental Reactor (ITER)**
Inertial Confinement Method (ICF)	Magnetic Confinement method (MCF)
Device called Hohlraum is used to achieve high temperature	Device called Tokomak is used.
Laser provide energy and thus achieve the high temperature and pressure	Powerful electric current and magnetic field is used to form plasma and achieve very high temperature and pressure
Done in a lab in USA	ITER is an international experiment and around 35 countries are participating in it.

Conclusion: Both ITER and NIF are working with the same goal – developing a fusion energy reactor. The NIF breakthrough is a giant leap forward, demonstrating the potential of fusion to revolutionize the energy landscape and contribute to a more sustainable future.

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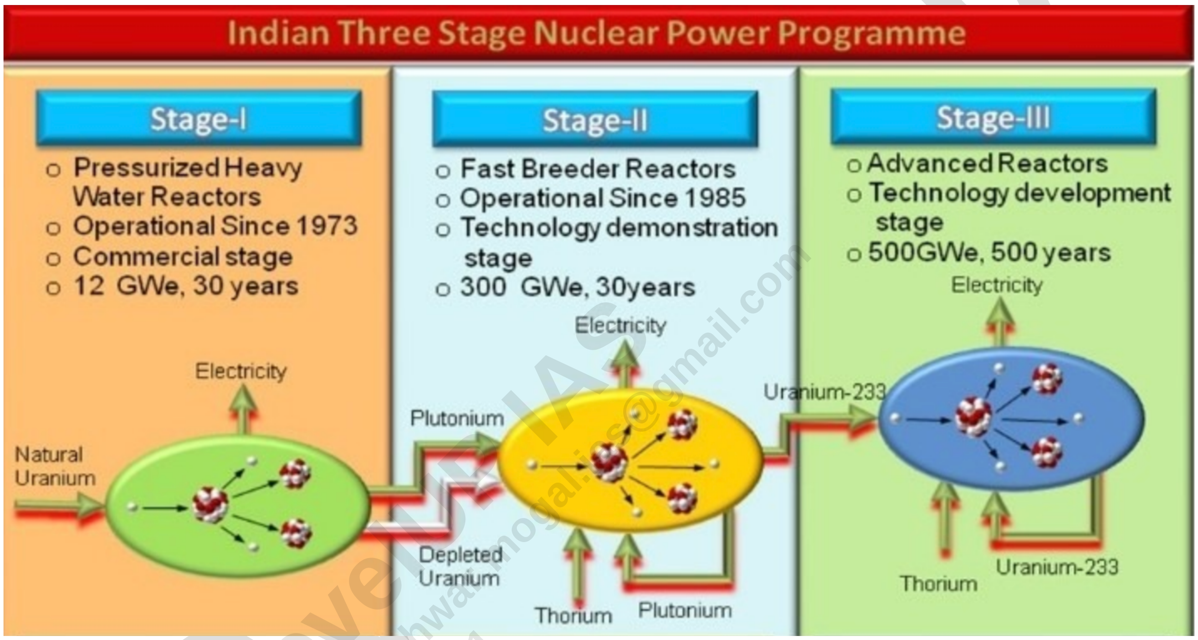
3) INDIA'S THREE-PHASE NUCLEAR POWER PROGRAM

- The three-stage nuclear power production program of India had been conceived by the **'father of Indian Nuclear Power Program' Dr Homi J Bhabha**, with the ultimate objective of utilizing the country's vast reserves of thorium-232.

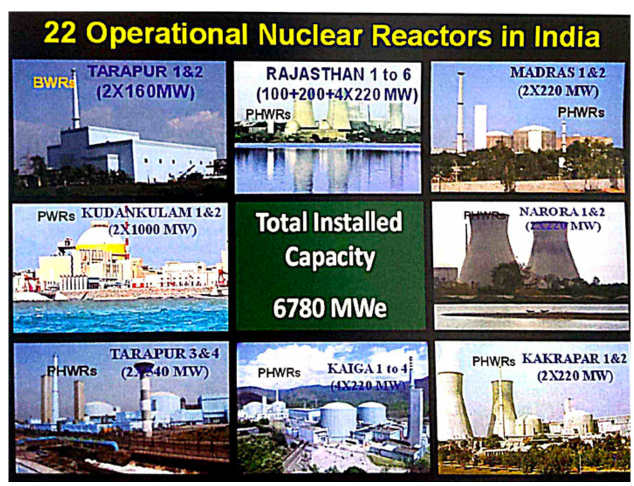
1. The first stage comprises setting up of **Heavy Water Reactors/Pressurized Heavy Water Reactors (PHWRs)** and associated fuel cycle facilities.

2. The second stage envisages setting up of **Fast Breeder Reactors (FBRs)** backed by reprocessing plants and plutonium-based fuels fabrication plants. Plutonium is produced by irradiation of U-238.

3. The third stage is based on **the thorium-232 -> Uranium 233 Cycle**, Uranium-233 is obtained by irradiation of Thorium.



- **Progress of the 3 Stages**
  - **The first stage** of Nuclear Power Programme is already in **commercial domain**. The Nuclear Power Corporation of India Ltd. (NPCIL), a public sector undertaking of DAE, is responsible for the design, construction, and operation of nuclear reactors. The company presently operates 23 reactors with a capacity of **7.7 GW**. The total operational capacity is expected to go to 22.4 GW by 2031



  - **The Second Stage** of Nuclear power generation programme is geared towards setting up the

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Fast Breeder Reactors. These reactors produce more fuel than they consume. The fast breeder program is in **technology demonstration stage**.

- **Features of the Prototype Fast Breeder Reactor (PFBR)**
  - **Fuel:** Plutonium Uranium Oxide (PuO_2 and UO_2)
  - **Coolant:** Liquid Sodium
  - **Liquid Sodium additional safety requirements**
    - Since sodium explodes if it comes in contact with water and burns when in contacts with air, additional safety requirements are needed to isolate the coolant from the environment.
    - Sodium also absorbs neutron to form Radioactive Na^{24} isotope.

  - **Advantages of FBR:**
    - They can ensure upto 60 times as much energy from the original Uranium compared with normal reactors.
    - Reduction in radioactive waste.
    - Safety -> closed fuel cycle would ensure safety
    - Energy security for India -> India plans third phase of its nuclear energy program on the success of FBR

- **The Third Stage:** of the Nuclear Power Programme is in **technology development stage**.
  - The ongoing development of 300 MWe Advanced Heavy Water Reactor (AHWR) at BARC aims at developing expertise for thorium utilization and demonstrating advanced safety concepts.
  - Thorium-based systems such as AHWR can be set up on commercial scale only after a large capacity based on fast breeder reactors, is built up.

  - Why Thorium based reactors are important for us
    i. **Abundance:** India has the world's third largest reserve of thorium.
    ii. **Less Enrichment requirement:** Thorium mining produces a single pure isotope, whereas the mixture of natural uranium isotope must be enriched to function.
    iii. **Superior Nuclear Properties:** Superior physical and nuclear properties
    iv. **Better Nuclear weapon resistant:** Better resistance to nuclear weapon proliferation
      - Weapon grade fissionable material (U-233) is harder to retrieve safely from a thorium reactor. It contains U-232, a strong source of gamma radiation that makes it difficult to work with. Further, its daughter product, thallium-208, is equally difficult to handle and easy to detect.
    v. Reduced plutonium and actinide production
    vi. They have miniscule long lived radioactive waste.

4) CIVIL LIABILITY FOR NUCLEAR DAMAGES (CLND) ACT, 2010

- The act which contains a speedy compensation mechanism for victims of nuclear accident.
- This act has been deemed responsible for Nuclear energy deadlock within the country. The two most contentious have been **Section 17(b) and Section (46)**

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- **Section 17(b) :** It contains provisions on **recourse liability on suppliers**. This allows a liable operator to recover compensation from a supplier in case the accident was caused by provisions of sub-standard services or defective or faulty equipment.
- **Section 46:** Potentially unlimited liability under this section. Section 46 provides that *nothing would prevent proceedings other than those which can be brought under the act, to be brought against the operator*. This is not uncommon as it allows criminal liability to be pursued where applicable.

5) NUCLEAR ENERGY AND ENERGY SECURITY

- **Introduction:**
    - Energy security means consistent availability of sufficient energy in various forms at affordable prices. When a country moves ahead on the path of development, it is necessary to utilize every energy resource available in the country.
    - Currently, nuclear energy makes up about 3% of India's energy sources
- **Advantages of Nuclear Energy:**
    a) **Least carbon footprint** (lesser than renewable energy): The threat of climate change and environmental pollution are likely to constraint the use of fossil fuels
    b) **Cost of nuclear powe**r: The cost of nuclear power plants is pretty competitive to other fossil versions
    c) **Quantity of waste generated** is also very less
    d) **Potential of self sufficiency** India has huge reserves of thorium which if properly utilized will reduce the dependency of India on foreign country
    e) **Depleting fossil fuels and import dependency**: India is currently drawing around 63% of its total energy from thermal sources. A significant part of this is imported.
    f) **Limitations of Renewable Energy**
        - Renewable energy are subject to vagaries of weather; they are land intensive; dependence on import technology; energy storage handicaps;
        - Renewable energy is inevitable and nuclear option should be retained as insurance.
- **Limitations**
    a) Safety concerns in light of recent disasters
    b) Nuclear waste disposal is a big concern
        - India still doesn't have a credible waste disposal policy.
    c) Potential of developing nuclear weapons
    d) Security concerns
        - Nuclear power plants can be favorite targets for terrorist organizations. If this happens it may cause irreversible damage to people living in the region and the ecosystem.
    e) India is dependent on other countries both for raw material and technology
        - Our future potentially depends on third stage of nuclear program.
    f) Ecological concerns
        - Nuclear plants are generally set near the coast as it requires a lot of water.
        - It is going to put pressure on coastline as India's western coastline is home to fragile ecology of western Ghats.
    g) Long gestation period

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- Till now only more than 20 plants are operational. There are long gestation periods which increases the cost of plants significantly.

h) More safeguards -> more costly
  - Post Fukushima disaster, the cost of per unit energy has gone up. This has led to concerns regarding the cost viability of nuclear power plants.

Way Forward

- As we know that India's total energy demand is expected to cross 800 GW by 2032, it is very important to utilize all possible options available and nuclear energy is one of the most important of those options.
- We need to develop a fledging domestic nuclear industry which will reduce our dependence as well as help us in reducing the gestation period of the plants.
- In light of the limitation's association with nuclear energy, stress should be laid on cautious development, safety precautions in operation and disposal of wastes. But development of nuclear energy can't be stonewalled in the light of such concerns.

6) USE OF NUCLEAR RADIATION TECHNOLOGY FOR PROVIDING BETTER QUALITY OF LIFE

1) Introduction

- Even before the use in nuclear bombs and nuclear energy, radiations were used in therapeutic and palliative for cancer cure in the early part of the 20th century.
- In the subsequent decades, nuclear fission made possible the harnessing of nuclear energy for electricity production.
- However, the peaceful uses of the atom have developed several other large-scale applications in agriculture, medicines and industrial sectors.

2) **Important fields in which radiation technology is being used**

1. **Health: Care to Cure**

   - Healthcare has grown into one of the most important peaceful uses of nuclear energy.
   - **Nuclear Medicine - Diagnosis**
     - Nuclear medicine is a medical speciality that uses **trace amounts** of radioactive substances (called **radio - pharmaceuticals**) in the diagnosis and treatment of wide range of diseases and conditions in a safe and painless way.
     - Radio pharmaceuticals can be administered by injection, inhalation, or orally and selectively localized and retained at sites of diseases. And thus, allow an image to be obtained of the loci using **gamma scintigraphy** or to deliver cytotoxic dose of radiation to specific disease sites without adversely affecting the surrounding normal tissues.
     - They help in identification of abnormalities in organ function even in early stages of a disease.
   - **Radiation Therapy**

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- A treatment that involve use of high-energy radiation either by using special machines or from radioactive substance. The aim is to impart specific amount of radiation at tumors or parts of the body to destroy the malignant cells.

1. **External Beam Radiation Therapy / teletherapy**
   - Radiation is delivered by using a machine outside the body.
   - A machine, either a **^{60}Co-teletherapy unit or linear accelerator** is used
   - It can be used to treat Breast Cancer, Bowel Cancer, Head and Neck Cancer and Lung Cancer.
   - **Bhabatron** is a teletherapy machine developed by BARC and has been installed in 50 cancer hospitals.
     - It is cheaper than any imported telecobalt machine.

2. **Internal Radiation Therapy or brachytherapy**
   - Radioactive material is placed in the body near cancer cells.
   - It makes it possible to treat a cancer with a larger dose of radiation that can't be given with external beam radiation therapy.
   - Tiny titanium encapsulated Iodine-125 seeds have been developed by BARC and have provided an avenue to treat eye cancer.

2. Food Security (1. Nuclear Agriculture 2. Food Preservation 3. Assessing the Quality of Food Products)

- Use of ionizing radiation based technologies provide **safe hygienic and economically viable** solutions to address issue of agricultural productivity

1. Nuclear Agriculture

- Ionizing radiation is being used by BARC to **induce mutation** in plant breeding, and around 50 varieties of different crops have been released to Indian farmers for commercial cultivation in the country.
  - e.g. groundnuts, mungbean, blackgram, pigeon pea, cowpea, mustard etc.
- Advantages
  - Higher yield
  - Earliness
  - Large seed size
  - Resistance to biotic and abiotic stress

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2. **Food Preservation - Produce and Preserve**

- Almost 30% of the food produced in India is lost due to spoilage because of pest attack, contamination and moulds infestation. These are encountered both during harvesting as well as post-harvest handling storage of the edible and cash crops.

- **Limitation of using pesticides**
    - Health hazards
    - Disturbance to ecology
    - Development of resistance in pest

- **Radiation Processing** can provide a viable, effective, and eco-friendly alternative to chemical fumigants and microbial decontamination, as the latter affect human health and environment adversely.

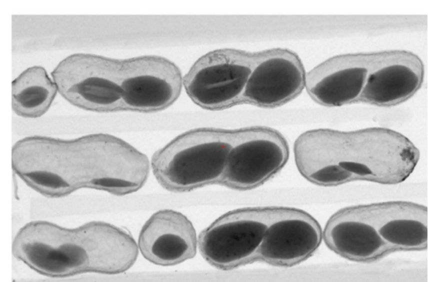
- There is an utmost need to adopt and integrate the irradiated foods into the country's supply chains and promote the widespread use of this technology to ensure food safety and security.

- **Advantages of using radiation processing**
    - Disinfestation of insects, pests in cereals, pulses and grain.
    - Microbial decontamination (hygienization) of dry species etc. for preservation/shelf life extension by applying pre-determined radiation doses.
    - Increasing the exportability of Indian food produce.
    - Elimination of parasites and pathogens of public health importance in food
    - Delay in ripening and senescence in fruits and vegetables
    - Inhibition of sprouting in tubers, bulbs and rhizomes

- **Radiation in no ways make production radioactive**.
    - Radiation therapy has been approved by WHO, IAEA, WTO, FSSAI etc.

Thus, as of Aug 2021, 26 Radiation Therapy plants are operational in the country.

3. Portable X-Ray imaging system can be very useful in grain value chains where the time needed to assess the economic value of grain by threshing or milling is a significant barrier.
    - For e.g., a team comprised of scientists from the International Crops Research Institute for the Semi-Arid Tropics (ICRISAT), Hyderabad and the Fraunhofer Development Center for X-Ray Technology (EZRT) in



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Erlangen, Germany have for the first time used **x-Ray radiography to determine key market-related traits of peanuts** while still inside the hull. (Sep 2022)

- X-Ray Radiography has the potential to be the right technology for in-field evaluation of farmers' produce which the International Committee for Food Value and Safety calls for.

3. **Energy Security - Nuclear is Clean and Green – already discussion**

4. **Societal Application**

    1. **Sludge Hygenisation - from waste to wealth** -> reduces spread of disease; protects environment; increased manufacture of manure etc.

5. **Hydrogel - Healing the wound**

    - The process was developed by BARC scientists and technologically has been transferred for commercial purpose.
    - Hydrogel is a thin transparent sheet of gel and is an excellent medical tool particularly useful for burn and injury dressings.
    - **Production**
        - It is prepared by cross linking molecules of hydrophilic polymers like PVA either chemically or by Gamma/Electron beam irradiation.
        - A 3D network of gel like structure is formed which holds large quantities of water. Gamma Irradiation achieves gel formation and sterilization in one step.

6. **Water Resources**

    1. **Isotope Hydrology techniques**
        - Isotope hydrology techniques enable **accurate tracing** of measurement of the extent of new and renewable underground water resources at various locations.
        - The data obtained is used towards resource planning and sustainable management of water resources.
    2. **Measuring contaminants in water**
        - BARC has developed low cost and user-friendly kits for measurement of contaminants in water.

7. **Industrial Applications**

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1. **Radiation Sterilization of Medical Products**

2. **Radiography**

   - Radioisotopes which emit gamma rays are more portable than x-ray machines, and may give higher-energy radiation, which can be used to check welds of new gas and oil pipeline systems, with the radioactive source being placed inside the pipe and the film outside the weld.

   - Radiography can also be used to gauge the thickness and density of materials or locate components that are not visible to other means.

7) IMPORTANT SCIENTISTS AND THEIR CONTRIBUTIONS: HOMI JEHANGIR BHABHA

- He is an Indian born nuclear physicist who made important contributions to quantum theory and cosmic radiations.

- For his contributions in Nuclear Science and Technology in India, he is also known as the "Father of India's Nuclear Science Program".
  - He returned to India from England in 1939 to join Indian Institute of Science, where he founded the Cosmic Ray Research Unit.
  - He understood the importance of **atomic energy** and wrote a letter to Prime Minister Nehru for promotion of the field. He established a **Nuclear Research Centre in Trombay in 1954** which later came to be known as **Bhabha Atomic Research Centre**.
  - With the help of J.R.D. Tata, he played an instrumental role in the establishment of the **Tata Institute of Fundamental Research** in Mumbai.
  - He **headed India's nuclear program** till his death in 1966.

  For his contributions, he was awarded **Adams Prize** (in 1942) and **Padma Bhushan** (1954). He was also nominated for Nobel Prize for Physics in 1951 and 1953-56.

8) RADIOCARBON DATING

- Radiocarbon dating is a method by which age of an object is determined using radiocarbon, a name for the isotope Carbon-14.
- **How does radiocarbon dating work?**
  - **When an organism is alive**, it constantly exchanges carbon with its surrounding by breathing, consuming food, defecating, shedding skin etc. Through these activities, carbon-14 is **both lost and replenished** in the body, so its **concentration in the body is nearly constant and in equilibrium with its surrounding**.
  - **When the living organism dies**, the C-14 is not replenished and it begins to **reduce due to radioactive decay**.
  - **Radiocarbon dating** dates an object by measuring amount of C-14 left, which scientists can use to calculate how long ago the body expired.

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- **Note:** Since carbon-14 decays with a half-life of around **5,730 years**, its presence can be used to date samples that are **around 60 millennia old (**i.e. 60,000 years old**)**. Beyond that, the concentration of carbon-14 in the sample would have declined by **more than 99%**.

- **Tools of Radiocarbon dating**:
    - **Geiger Counter** was used in 1940s when radiocarbon dating began. It consists of a **Geiger-Muller tube** connected to some electronics that interpret and display signals.
    - **Today**, more sophisticated devices are used. For e.g., one of the most sensitive dating setups uses **accelerator mass spectrometry (AMS),** which can work with organic samples as little as 50 mg.
- **Impact of Radio-carbon dating on science and technology**:
    - It was the first objective dating method to give numerical date to organic matter. Its impact on the field of archaeology and geology have come to be known as "**radiocarbon revolution"**.

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3. COMPUTER, IT AND ELECTRONICS

1) 5G

- 5G refers to **the fifth generation of cellular wireless communication technologies**.

- **Key features of 5G technology are**:
  - **Higher speed**; **Lower latency; Greater network stability**.
  - **Device Intelligence:** Unlike 4G, 5G has the capability to **differentiate between fixed and mobile devices**. It uses **cognitive radio techniques** to identify each device and offer the most appropriate delivery channel. This will allow a much more customized internet connection – according to device capability and local reception environment.
  - **Other technical features of 5G**
    - 5G will use **higher frequencies** of wireless spectrum (~ **30 GHz to 300 GHz**) range when compared to 4G which uses frequencies below 6 GHz.
      - **Higher Frequency** -> Huge quantity of data; Shorter Wavelength -> smaller antenna sizes.
    - These frequencies are **highly directional** and thus can be used right next to other wireless signals without causing interference.
    - Building on the multiplexing technology of its predecessor, 5G ushers in a new standard called **5G New Radio (NR),** which uses the best capabilities of LTE. **5G NR** will enable **increased energy savings for connected** devices and enhance connectivity.
    - Several hundreds of thousands of simultaneous connections for wireless sensors
    - Spectral efficiency significantly enhanced compared to 4G
    - Improved Coverage
    - Enhanced signal efficiency

- The **Ultra-fast speed of 5G indeed holds the promise of revolutionary changes in various sectors**:
  - **Upgraded Mobile Services**
  - **Economic Benefits**
    - » According to A.J. Paulraj Committee report, the overall impact of 5G on Indian economy can be upto $1 trillion by 2035. It will help business especially those that operate outside the reach of broadband networks or suffer from slow fixed line services.
    - » 5G would shape Fourth Industrial Revolution, or Industry 4.0
  - **Fast Rural connectivity**: Cost of putting up mobile data will be lower than fiber optics cable, 5G will help rural areas get faster connectivity. It will have huge positive impact on health, education, and other services in rural areas.
    - » Further, with IOT precision agriculture becomes more effective.
  - **Better Health services:** Telemedicine will be possible; Remote surgeries more effective due to lower latency.
  - **Ed-tech** sector will be able to reach more people and provide education in nooks and corners of the country.
  - **Better disaster management**: **->** Faster image transfers by drones – Faster Processing.

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- **Driverless cars** – large number of sensors in these cars will generate massive amount of data that will need to be communicated to other vehicles.
    - **Internet of Things (IoT):** 5G is considered backbone of IoT due to its high data rate, reduced end-to-end latency and improved coverage.
  - **Better Energy management** -> A smart network identifies the grid connectivity issues and takes appropriate action in real time by analyzing the data that originates from the two-way communication infrastructure.
  - **Better Law and Order Services** -> 5G adoption would ensure the best performance of police devices such as body cams, facial recognition technology, automatic number-plate recognition, drones and CCTVs.
  - **Other potential uses**
    - **Download** time for a high-definition full-length movie will be seconds, not minutes
    - It will allow **simultaneous language translation** between people attending teleconference.

- **Concerns which 5G may create:**
  - **5G and Cyber Threat Landscape**: Deploying 5G when we have a shaky cyber security foundation is like erecting a structure on soft sand.
    - Previous networks were hardware based. So, India could practice cyber hygiene. But **5G is a software-defined digital routing.** This makes it susceptible to cyber threats such as botnet attacks, man-in-the-middle attacks, and distributed Denial of Service (DDoS) overloads.
    - Further, 5G network will bring about a **wider proliferation of IoT-enabled devices**. This magnifies the threat canvas, as these devices will offer new malware and botnet distribution vectors.
  - **Geopolitical issues**: **Emerging threats from China and Countering it**: China has taken a lead in the development of 5G, and **Chinese telecom companies are aggressively penetrating new markets** by commercializing the technology and offering it at cheaper rates than their competitors. This has raised concerns that China may be strategically pushing these companies to capture global markets and therefore, may be **establishing an eavesdropping network**.
  - **Criminal elements** will also have all the benefits like faster and real time coordination.
  - **Privacy Concerns**: Unlike 4G, networks running on 5G have a much smaller area of coverage. This can allow **precise location tracking of mobile phone or internet users** inside and outside, potentially impacting their privacy.
  - **Other general issues**: Poor infrastructure in India; Only two major players left – Airtel and Jio – Reduced competition etc.

- **Way Forward:**
  - **Focus on cyber security:**
    - Updated latest technology.
    - Organizations connecting to 5G network must be cognizant of the evolving threat landscape and adopt security protocols accordingly.
    - A critical component of resilience will be awareness of end-users. Their **cyber hygiene** – their understanding of safe practices in cyber space – can help them better tackle the threats and protect themselves.
  - **International collaboration with** like-minded countries to deal with China's rise in 5G technology.

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- » **Fight new age 5G Crimes:**
  - **First**, the **police need to be trained** so that they recognize new 5G enabled crimes.
  - **Second**, **training programs** focused on 5G based crimes must be developed. This includes identifying potential scenarios for new types of crimes and their prevention.
  - **Third**, government and telecom companies may think of setting up **a 5G crime monitoring task force to monitor and identify new crimes** and develop counter measures.
  - **Fourth**, bring **changes in laws/rules/ regulations to prevent these kinds of crimes**.
- » **Provide effective regulatory framework** to allow fast track growth of 5G technology which will play a crucial role in economic growth of the country.

- **Conclusion**: To sum up, 5G offers new opportunities for digitalization and development, but the technology and network are not secure by design. Therefore, India must have a cyber resilience plan in place.

2) 6G

- Successor of 5G
- **Frequency Bands** – 95 GHz to 3 THz
  - » It seeks to use Tera Hz band frequency which is still unutilized. Tera Hz band fall between infrared and microwaves. Though the waves have very small wavelength, there is a huge amount of free spectrum which would allow us very fast data rates.
- Data rate – Upto 1 TBPS (100 times faster than 5G)
- Latency < 1 milli seconds
- **Application and Advantages** (Similar points as 5G)
- **6G also envisions to enable new applications** such as **holographic communication**, brain-computer interface, quantum internet, and artificial intelligence.
- **Challenges** for India:
  - » Low R&D investments
  - » Terahertz communication are blocked easily by barriers and signal also attenuates easily

A) BHARAT 6G ALLIANCE (B6GA)

- **Why in news?**
  - » DoT launches Bharat 6G Alliance to drive innovation and collaboration in Next-Generation Wireless Technology (July 2023)

- **Details** about **B6GA**:
  - » It is a **collaborative platform** consisting of public and private companies, academia, research institutions and standards development organization. It will forge coalition and synergies with other 6G Global Alliances, fostering international collaboration and knowledge network.

  - » The **primary objective** of the B6GA is to facilitate market access for Indian telecom technology products and services, enabling the country to emerge as a **global leader in 6G technology**.

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» It aims to bring together Indian startups, companies, and the manufacturing ecosystem to establish consortia that drive the design, development and deployment of 6G technologies in India.

» It also focuses upon accelerating standard related patent creation within the country and actively contributing to international standardization organizations such as 3GPP and ITU.

B) BHARAT 6G MISSION

- Aim of 6G service rollout by **2030.**
- India has also launched a development test bed.
- **More about the Vision document**
  - » **Prepared** by the Technology Innovation Group on 6G (TIG) which was set up by Department of Telecommunication in 2021.
  - » **Mission divided into two phases**:
    - **Phase 1** (2023-2025): Ideation phase – understand various potentials and risks; test proof the concept
    - **Phase 2** (2025-2030): Delivering the potential technology solution
  - » **Constitution of an apex body** to oversee the mission and approve the budget of the mission

- **Significance** of the document:
  - » Assuming leadership in setting the 6G standards
  - » Not delaying adoption (as has happened in previous generations)
  - » Ensuring latest technology coming to India in the fastest way possible.

3) WEB BROWSERS

- **Definition:**
  - A web browser is software that allows you to find and view websites on the Internet. They translate code into the dynamic webpage that forms the backbone of our online experience.

- **How do Browsers work?**

  Modern web browsers have multiple core components, each of which is a complex technology in itself.

A) REQUEST AND RESPONSE

- When you enter a website's address the browser sends a request to a server, asking for the contents of the specific web browser you're interested in.
- The server then formulates a response containing the information (or data) required to construct the web pages. This response embarks on its journey back to your browser, carrying the digital blueprint for the page you requested.

B) DECONSTRUCTING THE RESPONSE

- The response from the server is an amalgam of various files. Typically, these files have information encoded in three languages: **HTML, CSS, and JavaScript**. Each set of information plays a pivotal role in shaping the final presentation of the web page.

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C) RENDERING

- With HTML, CSS and JavaScript in hand, a browser begins the process of rendering. This involves deciphering the HTML to understand the structural arrangement, applying CSS for stylistic finesse, and executive JS to infuse interactivity.

D) MANAGING DATA

- Browsers serve as adept custodians for your digital footprint, so they also implement instruments like **cookies** and **cache** to enhance your online experience.

E) SECURITY

- Web browsers use an array of security measures to protect your data as they fly between your computer to various servers, via the internet, and even when they're stored on your computer. They do this by using **encryption protocols**, such as **HTTPS**, to create secure tunnels for data exchange shielding the information from prying eyes.

- Browsers also use warning systems to alert you about potentially malicious websites, preventing inadvertent exposure to threats.

Future of Internet Browsers:

- As technology hurtles forward, web browsers evolve in tandem. They are embracing new technologies like **Web Assembly**, a format that enables near-native performance within the browser environment.
    - **Note**: WebAssembly is a type of code that can run on modern webbrowsers – it is low-level assembly-like language with a compact binary format that runs with near native performance and provides languages such C/C++ with a compilation target so that they can run on web. It is also designed to run along JavaScript, allowing both to work together.

- **Support for VR and AR** experience is also on the horizon, promising immersive online interactions.

- **Privacy features** are being bolstered, providing users a greater control over their digital footprint.

Conclusion:
Web browsers are the unsung heroes of our digital endeavors, translating code into the dynamic web pages that from the backbone of our online experiences.

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4) WEB 3.0

Background: Understanding Web 1.0 and Web 2.0

- **Web 1.0** is the world wide web or the internet that was invented in 1989. It became popular in 1993. The internet in the Web 1.0 was mostly static web pages. Here most of the users visited websites and read and interacted with the static material available there. It was a **closed environment** and users themselves couldn't create post content and reviews.

- **Web 2.0** started in some form by late 1990s. By 2004, most of the features of web 2.0 was available for implementation. Here websites were more dynamic where users could create content, post comment, write reviews etc. They could also upload photos and videos. Primarily, a social media kind of interaction is the differentiating trait of Web 2.0.

Concerns of Web 2.0:

» Most of the data on internet is owned and controlled by a few **behemoth companies**. It has created issues related to data privacy, data security and abuse of such data. It has kind of disappointed experts that the original purpose of internet has been distorted.

Web 3.0

Web3 or Web 3.0 is a term used to describe the **next phase of the internet.**

- It runs on the decentralized technology of blockchain and would be different from web 1.0 and web 2.0. Here, users have ownership stakes in platforms (unlike now where tech behemoths control everything). Here users will control their own data.

significance of web 3.0:

- Prevents monopoly over data.
- Promotes data privacy.
- Increase competition in fields like search engine businesses as control over content now restricted to just a few companies would end.
- New technology will give India an opportunity to innovate and develop.

Future of Web 3.0: Will it take off?

- It is still in its formative stage and only the future would tell if this would be accepted by wider internet users or not. **Tech honchos** like Elon Musk and Jack Dorsey don't see a future for Web3.

Conclusion: Web3 may or may not become dominant mode of internet, but it has definitely raised some relevant questions.

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5) BLOCKCHAIN TECHNOLOGY

Introduction

- Blockchain is an **incorruptible, decentralized, digital ledger** of transactions that can be programmed to record not just financial transactions but virtually anything of value. It was first used in the design and development of Bitcoin – Cryptocurrency in 2009 by **Satoshi Nakamoto**.

Data/Transactions stored in the blocks are **secured against tempering using cryptographic hash algorithm** and are validated and verified through consensus **(consensus protocol)** across nodes of blockchain network.

- **Positives/Advantages**
  - **Security -> Built in robustness ->** no single point of failure i.e. no centralized points of vulnerability that hackers can exploit.
  - **Trust -> Increased Transparency and incorruptibility**
  - **Reduces the role of intermediary.**
  - **Speeds up the process.**
  - **Lowers transaction cost**
  - **Applications in various sectors**

- **Applications (Current and Future Potential)**
  - **Economy and Finance** offers the strongest use cases for the technology.
    - **Financial transactions** are typically granted by third party and block chain could be used to automate the process, reducing overall costs, by cutting out the middleman with autonomous smart contract acting as trusted intermediaries between parties on the network.
    - **Faster, Cheaper settlements could** save billions of dollars from transaction costs while improving transparency.
    - **Stocks, mutual funds, bonds, and pensions** may one day be stored on blockchains as many financial organizations explore the technology.
  - **Automotive:** Consumers could use the blockchain application to manage the **fractional ownership** in autonomous cars.
  - **Public Ledger Information:** Many governments are looking to adopt this technology to store information about the citizens and census. A decentralized platform to safely store data regarding, birth, death crime etc. can contribute to effectively curbing fraudulent activities.
  - **Healthcare:** Patients encrypted information could be shared with multiple providers without the risk of privacy breaches.
  - **Smart Contracts:** Every agreement, every process, every task, and every payment would have **a digital record and signature** that could be identified, validated, stored, and shared. Intermediaries like lawyers, brokers, bankers might not be necessary.
  - **Secure File storage**
    - Distributing data throughout the network protects files from getting hacked or lost.
  - **Supply chain auditing**
  - **Anti-Money laundering and Know Your Customer (KYC)**
    - AML and KYC practices have a strong potential for being adapted to blockchain.
  - **Changing the nature of ownership of assets and managing titles**
    - Blockchains will make all kinds of record keeping more efficient. Property titles are a case in point. They tend to be susceptible to fraud, as well as costly and labour intensive.

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6) NATIONAL STRATEGY ON BLOCKCHAIN

- **Introduction**: **MEITY** released national strategy on blockchain for its adoption in government systems, especially e-governance services in Dec 2021.

- **Key highlights**
    - The National Strategy on Blockchain provides insights on the government's strategies and offers recommendations for creating digital platforms.
    - It quotes Gartner report and lays out the **potential of the use of blockchain**, which is expected to reach business value of $176 billion by 2025 and $1.3 trillion by 2030.

    - **Vision and Objective:**
        - **Create Trusted digital platforms** through shared blockchain infrastructure
        - **Promote R&D, HR, entrepreneurship, innovation etc. in Blockchain sector.**
        - Promote **India as a hub of blockchain** based technologies and make India a global leader in the sector.
        - Ensure state of art, high quality digital service delivery to citizens and businesses.

    - It has **identified 44 key areas** where **blockchains can be applied**, including transfer of land and property, managing digital certificates, pharmaceutical supply chain, e-notary services, e-voting, smart grid management and electronic health record management.

    - It has **taken into consideration the blockchain-based platforms operated by governme**nts in China, Brazil, the UAE, and Europe.

    - The ministry has also recommended the formation of a **National Blockchain Framework (NBF)** that can be utilized in areas such as health, agri, education, and finance.
        - It will have **three types of participants**:
            i. Confident user of the technology (application developers),
            ii. Provider or operator of technology (infrastructure and services, Blockchain as a service)
            iii. Complete technology stack builder (IP creator)
    - A dedicated team would be identified to hand-hold the implementers at various central/state government applications.

    - It says that **"geographically distributed national-level shared infrastructure"** is needed to "enable citizen services at large scale and enable cross domain application development".
    - The strategy seeks to "create a trusted digital platform through shared blockchain infrastructure" and promote "R&D, innovation, technology and application development" around the technology.
    - **Five key areas of weakness in the technology** – Scalability, Security, interoperability, data localization, and disposal of records.

Key challenges to such widespread adoptions are – Adoption of technology, Regulatory compliance, identification of suitable use cases, finding the right data format, and awareness and skill set.

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7) SUPERCOMPUTERS

- **Example Questions**: What is a Supercomputer? Discuss the key targets of India's National Super Computing Mission [150 worlds, 10 marks]

- **Intro**
    - A supercomputer is a **computer with a high-level computational capacity compared to a general-purpose computer** or Supercomputer is a computer with **great speed and memory**. They are usually thousands of time faster than ordinary personal computers made at that time.
    - As per the 62nd edition of TOP500 released in Nov 2023, following are the **most powerful supercomputers currently**:
        - **USA's Frontier** is the most powerful supercomputer in the world reaching **1194 petaflops** (1.194 Exaflops)
        - **USA's Aurora** system is at 2nd spot with a capacity of **585.34 PFlop/s**.
            - **Note:** Aurora is currently being commissioned and will reportedly exceed Frontier with a peak performance of 2 EFLops/s when finished.
        - **Eagle** (installed in the Microsoft Azure Cloud in the USA), is at 3rd Spot. This is the highest rank a cloud system has ever achieved. It has the capacity of **561.2 PFlop/s**.
        - **Fugaku (of Japan)** is now ranked 4th (it was ranked second till July 2023 and ranked one till Nov 2021). It's capacity is that of 441.02 PFlop/s.
        - **LUMI (of European Union, Finland)** is ranked 5th with a capacity of 379.70 PFlops.
    - **Uses:** Super computers are generally used for scientific and engineering applications that must **handle very large databases or do a great amount of computation** (or both). Some of the key areas where supercomputers contribute are:
        - Weather forecasting
        - Climate research (E.g. Pratyush at IITM, Pune)
        - Code-breaking
        - Genetic analysis
        - Oil and gas exploration – Seismic processing in the oil industry: Supercomputers help to detect and accelerate deeper geological insights.
        - Molecular modelling
        - Other jobs that need many calculations including engineering, product design, complex supply chain optimization (actually any kind of optimization), Bitcoin mining etc.

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8) SUPERCOMPUTING IN INDIA

- In India, Indigenous development of Supercomputers began in 1980s. India's first Supercomputer was Param 8,000 which was created in 1991.

- Currently, as per the 62nd edition of TOP500 released in Nov 2023, the most powerful supercomputer in India is **AIRAWAT – PSAI** which is ranked 75 with a total capacity of 13.17 Petaflops. Thus in **terms of supercomputing power** India is way behind the world leaders.

- **National Supercomputing Mission (NSM)**
    - A visionary program, launched in 2015, to enable India to leapfrog to the league of world class computing power nations.
    - The mission is jointly steered by DST and MEITy.
    - **Implemented by** Centre for Development of Advanced Computing (C-DAC); Indian Institute of Science (IISc), Bangalore.
    - **Super Computing Grid**: The mission envisages empowering our national academic and R&D institutions spread over the country by installing a vast supercomputing grid comprising of more than **70 high performance computing facilities**.
    - **Human Resource**: The mission also includes development of highly professional High-Performance Computing (HPC) aware human resource for meeting challenges of manpower scarcity in the sector.

- **Conclusion and Way Forward**:
    - Supercomputers are now using exa-scale systems which may reach its speed barrier soon. The scientists are already working on new mechanisms like quantum computing, optical computing etc. India should ensure more research in this field to ensure that future supercomputing doesn't remain a concern for it and it is able to walk along the advanced economies of the world in terms of computing power.

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9) QUANTUM COMPUTERS

- **Background**: Classical computers have enabled the information revolution that we are part of today. But these **classical computers can't do a number of things** including Optimization, Simulation of large molecules, factoring of large numbers etc. **But** Quantum computing may help us solve the above problems someday.

- **Quantum computers are based on the principle of quantum theory.** They gain enormous processing power due to the ability of quantum computer to perform task using all possible permutations simultaneously.

- **Quantum Computers** use **qubit** (Quantum bit). These qubits can take values 0 or 1 or any of the infinite **superpositions** between 0 and 1. When Qubits are in superposition, it has some probability of being in state 0 and some probability of being in state 1.
  - **Qubits** are usually made of things like electrons, photons or even a nucleus.

- **Quantum Supremacy:** It refers to quantum computers being able to solve a problem that a classical computer cannot. The term was coined by theoretical physicist **John Preskill** of the Caltech in 2012.
  - **Google** recently used a 53 Qubit processor (Sycamore) to generate a sequence of millions of numbers, that conform to an algorithm generated by google. A classical supercomputer checked some of these values and they were correct.
  - **Google's Quantum computer claimed** 'Supremacy' because it reportedly did the task in 200 seconds that would have apparently taken a supercomputer 10,000 years to complete.

- **Some Problems faced by Quantum Computing Sector:** While the above concept sounds promising, but there are still tremendous obstacles to be overcome.
  - **Interference:** During the computation phase of a quantum calculation, the slightest disturbance in the quantum system (a stray photon or a wave of EM radiation) causes the quantum computation to collapse, a process known as **Quantum Decoherence**.
  - **Error Corrections:** Because truly isolating the quantum system has proven so difficult, error correction systems for quantum computing have been developed.
  - **Output observance:** Observing the final output also risks corrupting the data.

- **The breakthroughs in the last 20 year** including the **quantum supremacy** achieved by Google have increased the chances of developing practical quantum computing mechanisms. However, it is not clear whether the practical application is less than a decade away or a hundred years into the future.

- **Examples of Quantum Computers**: While the idea governing quantum computers have been around since the 1990s, the actual machines have been around since 2011, most notably built by Canadian company **D-Wave systems**.

  - The recent Google's **53 qubit Quantum computer** is called **Sycamore**. Google is also spending billions and targets to build its own working quantum computer by 2029.

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» **IBM** plans to have a 1,000-qubit quantum computer. For now, IBM allows the use of its machines by those research organization, institutions etc which are part of its **quantum network**.

» **Microsoft** also offers companies access to quantum technologies via its **Azure Quantum Platform**.

- **Applications:** The potential that this technology offers are attracting tremendous interest from both the governments and the private sector.
  - » **Military Applications** include breaking of advanced encryption using brute force searches.
  - » **Advanced Cryptography:** Quantum uncertainties could be used to create private keys for encrypting messages to be sent from one place to another.
  - » **Climate Change and Weather Forecasting**
  - » **Faster Data analysis in industrial science applications** will enable faster solution to business problems in the era of big data.
  - » **Improved Optimization** for complex problems like **NP-hard problems**. This may lead to faster optimization of very large-scale problems involving complex network structures, computational biological science, and physical sciences.
  - » **Transform Healthcare and Medicine:** Drug Development and Discovery
  - » **Other civilian applications** include **DNA Modelling** and **complex material science analysis**.
  - » **Improved Machine Learning Outcomes** by enabling more efficient optimization of these algorithms so that ML capabilities become more efficient, accurate and fast.
  - » **Teleporting the information from one location to another** without physically transmitting the information. Entangling of quantum particles allow us to achieve this.

- **India and Quantum Computing:**
  - » There are no quantum computers in India yet.
  - » **Cabinet Approves Rs 6003 Crore National Quantum Mission** (April 2023)
  - » In Aug 2021, India launched **QSim** to aid Quantum Computing research in India.

- **Way Forward**:
  - » **Well-funded Research Centres of Excellence** – in the leading technological institutions. A long-term program by **DST** could possibly be introduced whereby strategic infrastructure and manpower training projects can be funded in the established technology engineering situations. This would need to cover both hardware and software to further develop a homegrown technology industry.
  - » **Clear and Sustained Policy and Governance**: Since quantum computers will deal with new levels of data and computing, it's important to create legal framework surrounding data management, data sharing, data privacy, information assurance, algorithmic governance, and transparency needs to evolve.
  - » **Manpower Skilling** is an important component of employability of the future workforce of India, and this may require policy interventions since most private organizations focus on exploiting immediate skill availability and project needs by compromising future skilling needs.
  - » **International Collaboration**: International funding agencies could develop joint project funding schemes whereby collaborations can be fostered to enable faster development in the space.
  - » **Encouraging Startups** focus on quantum computers: Through, government support via organizations like technology development boards, where grants can be given to startup ventures.

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- **Conclusion:** Quantum computing domain needs greater focus in the country through strategic investment in research, development, and training mechanisms. This may enable improved capability for leveraging and exploiting this domain for the benefit of citizens and the nation going forward.

10) INTERNET OF THINGS (IOT)

- **Introduction**
  - IoT is a network of physical objects embedded with sensors, software, and other technologies for connecting and exchanging data with other devices and systems via the internet.
  - **A thing** on the internet of Things, can be a person with a heart monitor implant, a farm animal with a biochip transponder, an automobile with a built-in-sensors to alert the driver when tire pressure is low - or any other natural or manmade object that can be **assigned an IP address** and provided with **the ability to transfer data** over a network.

- This is achieved by **sensors** and finally fabricated **micro-controllers**.

- **Advantages**
  - **Reduce waste, loss, and cost** -> by early detection of problems and taking corrective steps
  - We would know what things needed **replacing, repairing, or recalling** and whether they were fresh or past their best. This helps in increasing the **reliability** of a device.

- **Applications**
  - **Health Care Sector:** IoT can improve the reliability and performance of the life-critical system. For e.g., the IOT based devices can be used in combination with cardiac monitor to raise an alarm to the doctors in case of abnormality.
  - **Agriculture Sector**: IoT can be used to gather **live pedological data** that can be used by scientists to improve the yield of the land. It can also help in implementing **precision agriculture**.
  - **Transportation Sector**:
    - » **Early detection of wear and tear** (preventing accidents)
    - » Self-Driving Cars – will need IOT for real time decisions
    - » Traffic Management – real time traffic data -> better traffic management.
  - **Energy Management**
    - » Managing temperature in a Nuclear Power Plant (using sensors and IoT)
    - » Real time efficiency analysis of Solar Power panels.
  - **Research and Development**:
    - » E.g. – Recent development of wireless communication system for satellites by NASA through which Satellites can communicate with each other.
  - **Safety and Security**
    - » Real time tracking of criminals – using tagging and IoT.

- **Some Limitations of IoT**
  a. **High Initial cost of set up** -> Since IoT is based on expensive sensors
  b. **Increased cyber security concerns** -> with increased number of devices connected to internet
  Compatibility issues -> due to lack of the international standardization on IoT devices.

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11) BIG DATA

- **Intro**
  - » Big Data is a collection of data that is huge in volume (petabytes and exabytes of data) yet growing exponentially with time. It is a data with so large size and complexity that none of the traditional data management tools can store or process it efficiently. Big Data can be structured, semi-structured and Unstructured. But they generally have potential to be mined for information.

  - » **Examples of Big Data:**
    - ○ BSE which generates Gigabytes of data per day
    - ○ Social media – Around 500+ terabytes of new data get ingested into the database of social media site Facebook every day.
    - ○ Data from search engines (like Google, Bing etc.) and Online portals like Amazon.

- **Challenges** include capture, analysis, data curation, search, sharing, storage, transfer, visualization, querying, updating, and information privacy.

- Big data is **characterized by 3 Vs – Volume, Velocity and Variety**.

- **Advantages – Accuracy, Better Correlation**

- **Key areas where it can be used**
  - » Internet
  - » Finance
  - » Urban Informatics
  - » Business informatics
  - » Meteorology
  - » Genomics and healthcare
    Find new cures, optimize treatment, and even predict diseases before any physical symptoms appear
  - » Complex physical simulations
  - » Environment research
  - » Improve the performance of Individuals
    - ▫ (At sports, at home or work), where data from wearable sensors in equipment and wearable devices can be combined with video analytics to get insights that traditionally where impossible to achieve)

  - » Security Agencies
    - ▫ To prevent cyber attack
    - ▫ Detect credit card frauds
    - ▫ Foil terrorism
    - ▫ Even predict criminal activity

  - » Improve our homes, cities, and countries
    - ▫ Optimizing heating and lighting in our homes

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- Optimizing traffic flow in our cities
- Optimizing Energy Grid across the country

- **Relation between cloud computing and big data**
  - » Cloud computing is very important in BIG data analytics due to its application sharing and cost-effective properties

12) NOTE:

We will cover more topics of S&T in coming classes, but in next class we want to start with EB&CC. We will continue doing some S&T topics including AI, ChatGPT, IPR, Physics, Chemistry related update in the from of Current Affairs Updates

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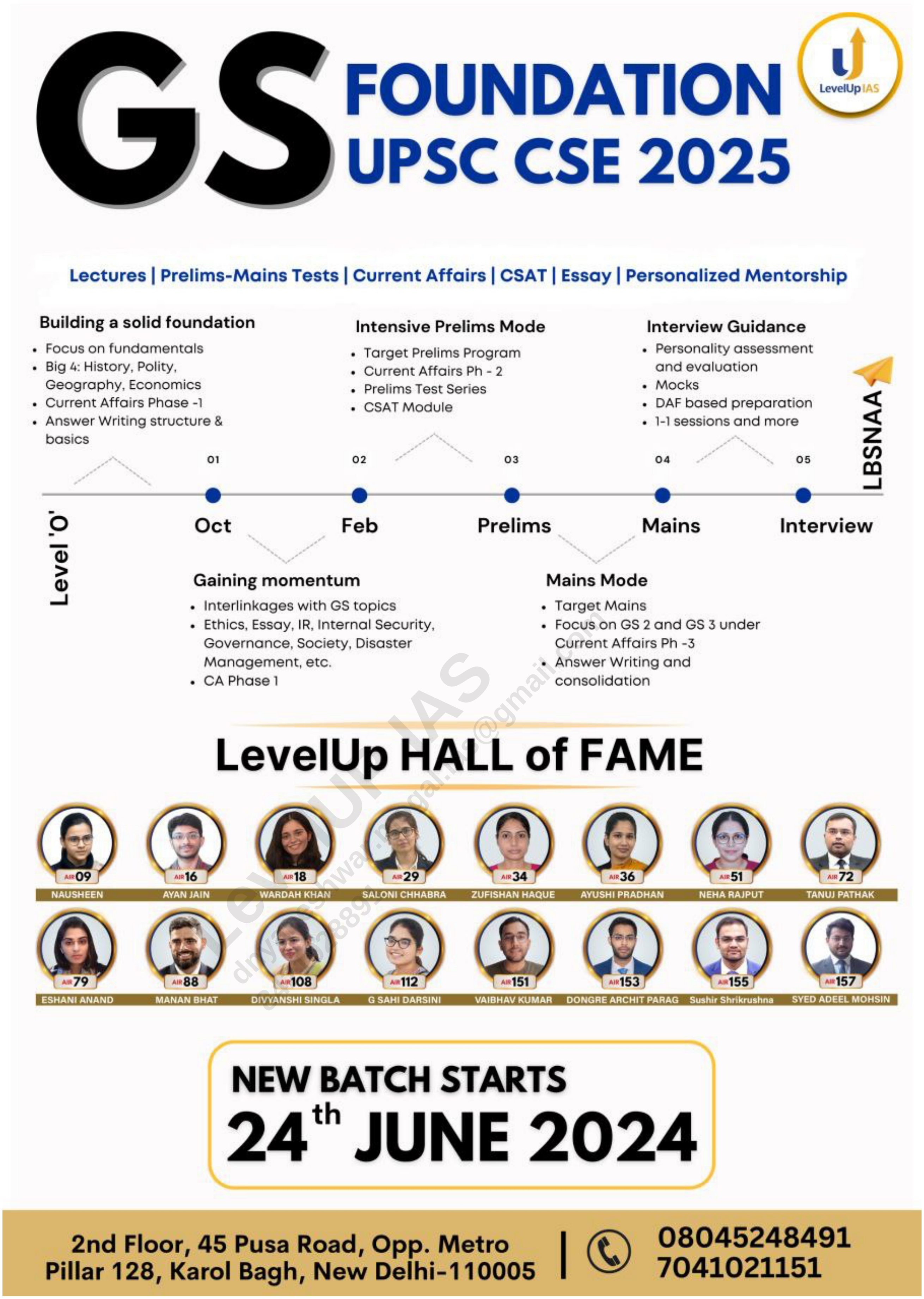
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Personality assessment and evaluation
Mocks
DAF based preparation
1-1 sessions and more
LBSNAA
01
02
03
04
05
Level 'O'
Oct
Feb
Prelims
Mains
Interview
Gaining momentum
Interlinkages with GS topics
Ethics, Essay, IR, Internal Security, Governance, Society, Disaster Management, etc.
CA Phase 1
Mains Mode
Target Mains
Focus on GS 2 and GS 3 under Current Affairs Ph -3
Answer Writing and consolidation
LevelUp HALL of FAME
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